

Data Mining Study of Pre-K Literacy in Memphis

Examining the Effects of Funding Source, Licensing Type, and Geographic District

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ABSTRACT

Early childhood literacy proficiency can be an important predictor for long-term academic success, yet large disparities persist across pre-k programs in Memphis, Tennessee. This study applies data mining to examine whether funding source, licensing type, program capacity, and geographic district are associated with differences in reading and writing proficiency across pre-k centers. Using a dataset from DataMidSouth, we conducted data preprocessing, visualization, correlation analysis, linear regression, and clustering to analyze program characteristics and academic outcomes. Results show that funding sources have a weak relationship with literacy proficiency, with regression and clustering indicating that funding labels alone do not meaningfully predict performance. In addition, although three major licensing categories (DOE, DHS, and Unlicensed) exist in the dataset, proficiency scores are reported only for DOE centers, limiting direct comparison across licensing types. In contrast, district level analysis reveals substantial geographic disparities in proficiency, with higher performing districts showing greater access to licensed and DOE supported programs. Program capacity also shows only a weak association with proficiency. These findings suggest that geographic and structural factors play a greater role in shaping early literacy outcomes than funding source or capacity alone, highlighting the importance of district focus policy and access to high quality pre-k programs.

1 Introduction

The quality of early childhood education is important in determining long-term academic, social, and economic outcomes. Identifying factors that influence early proficiency rates can help guide funding decisions, policy choice, and resource allocation for Pre-K programs across the Mid-South region. In this study, we analyzed four main characteristics of Mid-South Pre-K centers – total capacity, funding source, licensing type, and geographic district – to determine which factors are most strongly associated with student proficiency.

Existing educational research has shown that socioeconomic conditions, neighborhood characteristics, and institutional quality play a significant role in shaping

early learning outcomes ((e.g., Duncan & Magnuson, 2013; Chetty et al., 2014). However, little work has specifically analyzed these factors at the level of Pre-K centers within Shelby County and the surrounding Mid-South areas. Our dataset provided an opportunity to investigate these relationships quantitatively.

This report addresses the following research questions:

- Does the total capacity of a Pre-K center correlate with its percent proficiency?
- Does the funding source of a Pre-K center correlate with its percent proficiency?
- Do Commission Districts differ significantly in their proficiency outcomes?
- Do licensing classifications – such as DOE (Department of Education) and DHS (Department of Human Services) — have an effect on proficiency performance?

2 Materials

2.1 Dataset

The primary dataset used in this study was obtained from the DataMidSouth Open Data Portal, specifically from the Pre-K Capacity 2024 data release. The dataset includes over 300 records across the Mid-South region and contains attributes such as Pre-K Center Name, License Type, 3-Year-Old Capacity, 4-Year-Old Capacity, Total Capacity, Valid Tests, Proficient Tests, Percent Proficient, Address, Zip Code, Latitude, Longitude, Commission District, City Council District, and Super District.

2.2 Data Cleaning and Preprocessing

Several steps were required before analysis:

- CSV parsing: The file contained inconsistent delimiters and malformed lines, which required cleaning to ensure proper import into Python
- Type conversions: Numeric values such as Total Capacity and Percent Proficient were converted from strings to numeric data types.

- Handling missing values: Records containing missing proficiency data were removed from analyses. Only centers with complete values for Total Capacity and Percent Proficient were retained, resulting in a final dataset of 194 centers.

3 Methods

3.1 Regression Analysis (Capacity and Proficiency)

A linear regression analysis was performed to determine the relationship between total capacity and percent proficient. The expectation was that larger centers might have more resources or greater instructional quality, potentially leading to better proficiency outcomes.

3.2 K-Means Clustering (Capacity and Proficiency)

K-means cluster, with (k=3), was applied to total capacity and percent proficient. The algorithm was iterated 20 times to ensure stable cluster formation. The goal was to determine whether clusters will form based on capacity size or proficiency rates.

3.3 Commission District Level Comparison

A box plot was used to visualize the difference in proficiency across Commission Districts. A one-way ANOVA test was conducted to test whether district means differed significantly. This method allowed us to assess geographic disparities and identify district-level patterns.

3.4 Funding Source Boxplot Analysis

A boxplot was constructed to visualize Percent Proficient across different funding sources. The boxplot allowed for direct visual comparison of variability and central tendency across funding sources.

3.5 Funding Source Regression Analysis

A linear regression analysis applied to funding source and percent proficient. Because funding source is a categorical variable, each funding category was converted into a numeric variable. This method was selected to determine whether funding sources could be used as a predictor of literacy performance across Pre-K centers.

3.6 Funding Source K-Means Clustering

K-means with 3 clusters was also applied to funding sources and percent proficient. Determining whether Pre-K centers cluster based on funding structure and proficiency

levels, or whether funding types are distributed evenly across performance groups.

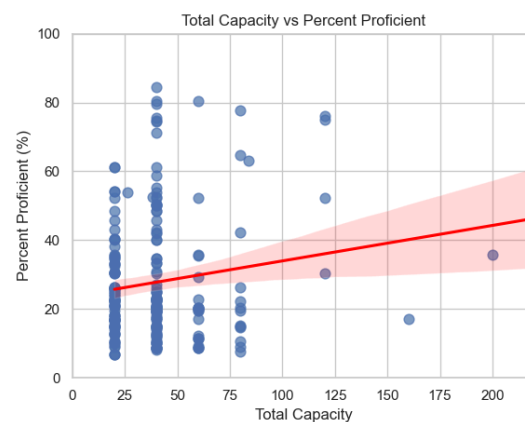
3.7 Licensing Classification Analysis

To evaluate whether licensing structure influences literacy proficiency, centers were categorized into licensing groups: DOE (Department of Education), DHS (Department of Human Services), and Unlicensed. However, during preprocessing it was found that only DOE centers had reported proficiency data. As a result, no statistical comparison across licensing groups could be performed. This licensing analysis was retained to document data limitations and assessment coverage differences across program types.

4 Results

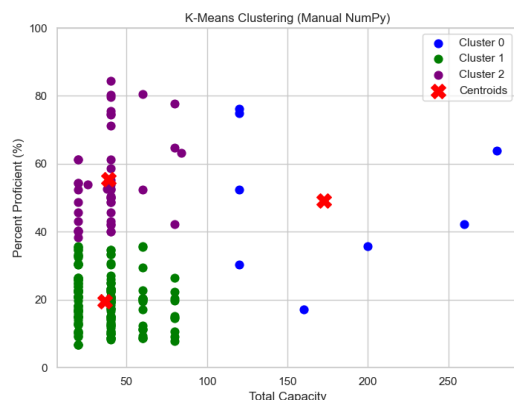
4.1 Total Capacity Does Not Predict Proficiency

Regression analysis shows no clear relationship between total capacity and percent proficient. The regression slope was near zero and the correlation coefficient was close to 0. These results indicate that capacity is not associated with proficiency rates.



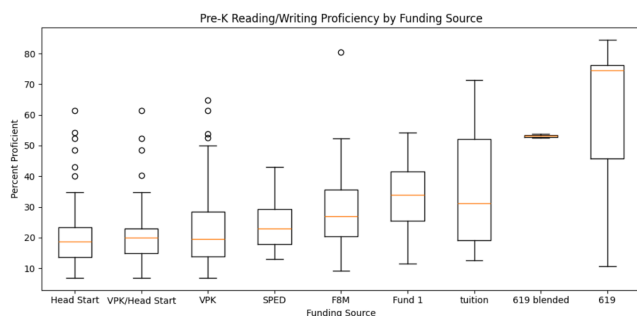
4.2 K-Means Confirms Weak Relationship

K-means clustering did not reveal distinct capacity-based groups. Instead, clusters primarily formed around proficiency levels, instead of capacity. Since both high-performing centers and low-performing centers appeared across a wide range of capacities, it indicates that capacity is not a predictor for proficiency performance.

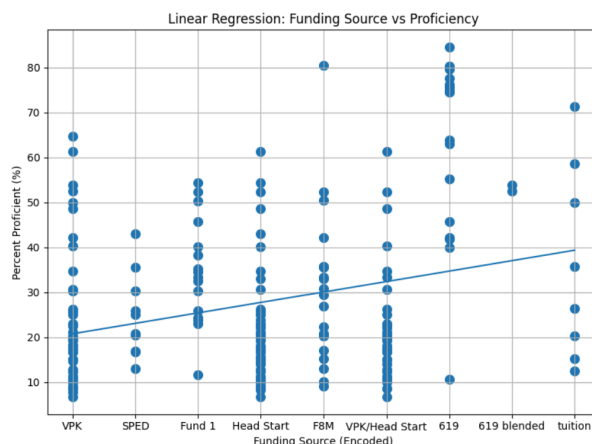


4.3 Funding Sources Does Not Predict Proficiency

Boxplot analysis revealed overlap in Percent Proficiency across all funding categories, including VPK, Head Start, F&M, 619, tuition-based, and blended programs. While some funding sources have slightly higher median proficiency values, no category consistently outperformed the others. Each funding group contained both high- and low-performing centers, indicating that proficiency varies more within funding categories than between them.

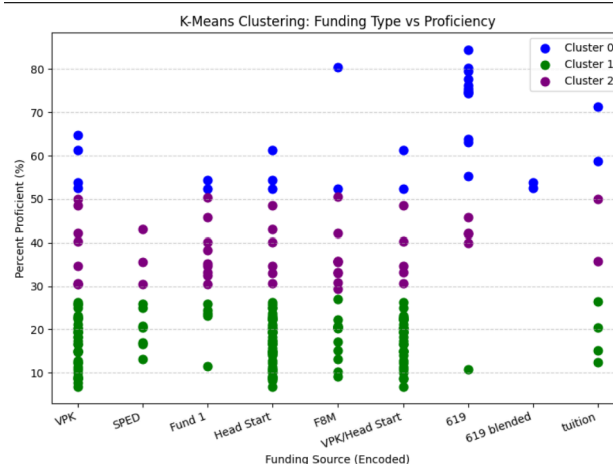


Correlation and regression analyses further supported this observation. The regression slope was near zero and the R^2 value was very small, indicating that funding source explains almost none of the variation in literacy proficiency. These results demonstrate that funding source alone is not a meaningful predictor of proficiency performance.



4.4 K-Means Confirms Weak Relationship

K-means clustering with $k = 3$ was applied using encoded funding source and Percent Proficient as features. The resulting clusters formed primarily around proficiency level (low, medium, and high) rather than funding structure. Each cluster contained a mixture of funding types, including VPK, Head Start, 619, tuition-based, and blended programs.

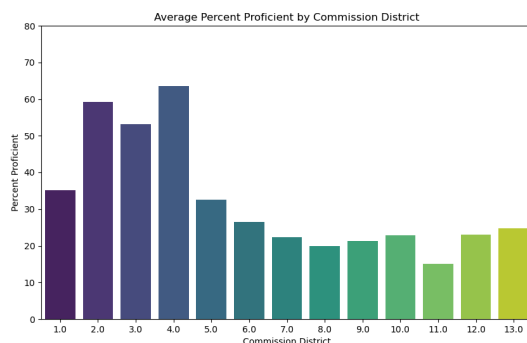
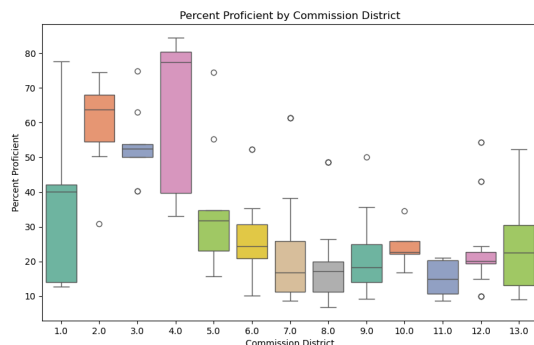


High-performing and low-performing centers were distributed across nearly all funding categories. This confirms that centers with similar funding sources do not consistently exhibit similar proficiency outcomes. These clustering results reinforce the findings from the regression and boxplot analyses that funding source does not determine literacy performance.

4.5 District Differences Are Large and Significant

Districts 1-4 revealed a much higher average proficiency percentages, while Districts 5-13 performed significantly worse on average. The ANOVA yielded an F-statistics of 14.21, indicating a strong variation in proficiency between each district. The extremely small p-value (2.64×10^{-21})

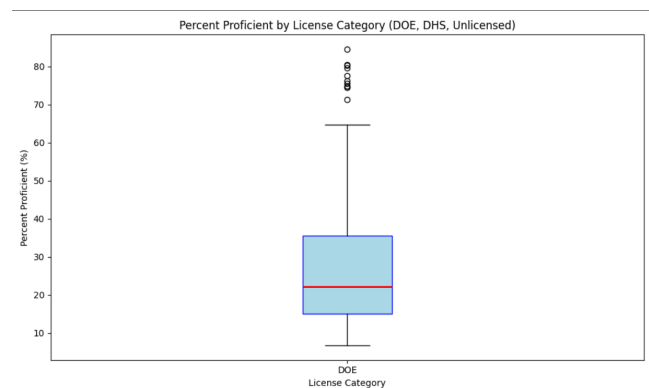
confirms that these differences are significant and unlikely due to random chance. Therefore, a geographic district is a strong predictor of proficiency.



In addition, a bar chart was used to visualize average proficiency rate by Commission District. The graph clearly reinforces the same pattern observed in the box plot. Districts 1-4 reveal much higher proficiency than Districts 5-13. The difference in heights of the bars provide a clear visualization of the sharp divide between the higher-performing districts and the lower-performing districts. This chart aligns with the results from the ANOVA test and confirms that districts influence proficiency outcomes.

4.6 Licensing Testing

Although the dataset includes three licensing types: DOE (Department of Education), DHS (Department of Human Services), and Unlicensed, proficiency scores were reported only for DOE centers. All DHS and Unlicensed centers contained missing Percent Proficient values and were removed during preprocessing. As a result, no statistical comparison could be performed across licensing categories.



4.7 Socioeconomic Context Explains the Disparities

To better understand the substantial differences in proficiency across Commission Districts, we examined socioeconomic data for the neighborhoods represented in the dataset. Districts 1-4, which showed the highest proficiency rates, include communities such as Germantown, Collierville, East Memphis, Cordova, and Bartlett. According to the Statistical Atlas of the United States, these regions have median household incomes ranging from approximately \$75,000 to over \$120,000, along with higher tax revenues.

In contrast, District 5-13, which shows the lowest proficiency outcomes, includes communities like Frayser, North Memphis, South Memphis, and Whitehaven. The median household incomes in these regions range from \$20,000 to \$40,000, along with poverty rates and greater housing instability.

These socioeconomic patterns align with the disparity patterns observed in box plot, bar chart, and ANOVA results.

5 Conclusion

Our analysis shows that total capacity does not have a meaningful correlation with proficiency rates, indicating that center size is not a reliable predictor of early learning outcomes. Similarly, funding source does not independently predict literacy performance, as proficiency varies widely within each funding category and shows only weak statistical associations.

In contrast, geographic districts emerge as a strong and significant predictor of proficiency, with Districts 1-4 demonstrating substantially higher average performance than Districts 5-13. The absence of testing data among

DHS and Unlicensed centers further highlights systemic gaps in assessment coverage and limits the ability to compare instructional quality across all program types.

Together, these findings suggest that structural and geographic factors play a more influential role in shaping early literacy outcomes than capacity, funding sources, or licensing types alone. Future work should prioritize expanding assessment coverage across all program types and examining teacher qualifications, curriculum quality, and classroom level instructional practices to better understand the drivers of early learning success.

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